Beyond Fake or Real: Evidence-Grounded, Explainable Five-Class News Verification with Temporal Context Checking

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Abstract—Most "fake news" detectors are binary text classifiers that learn how fabricated articles are written, not whether their claims are true; they cannot explain a verdict, describe partly wrong or outdated items, or abstain. We present an open-source pipeline that runs on a student laptop. It extracts check-worthy claims, retrieves evidence from free web sources and an offline index of fact-checked statements, labels the stance of every passage with an NLI cross-encoder, checks dates for old news presented as new, and fuses these signals with transparent rules into one of five verdicts—Real, Fake, Partially True, Misleading or Unverifiable—with a confidence score, dated evidence and an explanation. A fine-tuned DistilBERT style classifier reaches 0.9898 test macro-F1 on WELFake but may only adjust the confidence, because TF-IDF classifiers lose 0.29–0.44 macro-F1 when moved to a disjoint corpus. On 65 held-out fact-checked claims from 2025–26 the full system reaches a five-class macro-F1 of 0.1667 online and 0.0343 offline, far below our 0.55 target, mainly because it abstains too often. We also show that an earlier run leaked each claim's own fact-check page into the evidence, which inflated macro-F1 from 0.1260 to 0.1854. Code, data splits and claim sets are released.

Index Terms—fake news detection, fact verification, evidence retrieval, stance detection, natural language inference, explainable AI, temporal misinformation

I. Introduction

News now reaches most people through feeds and messaging groups rather than through editors, and the damage done by what slips through is well documented. Tracing roughly 126,000 story cascades on Twitter over eleven years, Vosoughi et al. found that false stories were about 70 % more likely to be retweeted than true ones and that the truth needed roughly six times longer to reach 1,500 people [1]. India, the setting of this work, offers its own examples: Indian fact-checkers routinely find genuine footage of earlier floods, riots and accidents re-captioned as "breaking" news for a new event—in one analysis of 125 fact-checked Indian COVID-19 items, nearly half paired text with video and 94 % originated on social platforms [2]. This last pattern—true material presented in the wrong time or place—is one that most detectors are not built to see.

Manual fact-checking is accurate but slow. Most academic work on automatic detection has therefore taken a shortcut: treat the task as binary text classification and let a model learn the surface cues of fabricated writing. Such classifiers report accuracies above 95 % on benchmark corpora [3], [4], but the numbers hide three weaknesses. First, they learn style and dataset artefacts rather than facts: a corpus whose genuine articles all come from one wire service can be separated almost perfectly by boilerplate alone [5], and the same models lose much of their accuracy on a different corpus. Second, a binary label cannot express the shapes real misinformation takes—a true event with a wrong number, a real quotation attached to the wrong occasion, a 2018 video captioned as today's news. Third, a classifier always produces an answer; it cannot say that the evidence is insufficient, which for hyper-local or very recent claims is the only honest response.

Problem statement. Given a news item—a headline, an article text or a URL—automatically determine, with supporting evidence, dates and a human-readable explanation, whether its central claims are real, fake, partially true, misleading or unverifiable, using only free tools and hardware available to students.

We address this problem by moving the decision from the classifier to the evidence: the system selects the check-worthy sentences of an item, retrieves passages from free web search, Wikipedia, the Google Fact Check Tools API and an offline index of fact-checked statements, labels each

passage as supporting, refuting or neutral, compares the dates involved, and applies a small, inspectable set of rules. A style-based classifier is retained, but it may only adjust the confidence or break a single-source tie; it never outputs Fake or Real on its own when evidence exists. The contributions of this paper are:

- An evidence-grounded five-class verdict scheme with abstention. We define Real, Fake, Partially True, Misleading and Unverifiable operationally, map LIAR's six PolitiFact grades onto them, and reach each verdict through a published decision table whose confidence reflects evidence agreement, coverage and source reliability, so every output carries the rule that fired, the evidence used and an explicit uncertainty statement (objective O6).
- A laptop-scale claim-to-verdict pipeline. Claim identification, evidence retrieval from free sources with an offline FAISS fallback, NLI stance detection with a small cross-encoder, and a temporal/context check that flags old news presented as new and entity–date mismatches, all running on CPU within seconds offline (objectives O3–O5).
- An explainable content classifier evaluated honestly. A fine-tuned DistilBERT model with LIME token attributions serves as the style signal; we report in-domain results on WELFake, a cross-dataset transfer and leakage audit against ISOT, and an ablation of how much the classifier adds to the evidence-based verdict (objective O1).

The rest of the paper is organised as follows. Section II surveys related work and states the gap; Sections III and IV describe the datasets and the pipeline with its fusion rules; Section V reports results, Section VI compares the system with existing ones and lists its limitations, and Section VII concludes.

II. Related Work

We group prior work into five themes that correspond to the components of our system; Table I summarises the most relevant papers along the dimensions that matter to us.

A. Datasets and Benchmarks

LIAR [6] collects 12.8k short political statements rated by PolitiFact on a six-point scale from pants-on-fire to true. Its best text-and-metadata model reached only about 27 % six-way accuracy, which remains a useful reminder that fine-grained truthfulness cannot be read off the wording of a sentence. ISOT [5] pairs 21k Reuters articles with 23k articles from sites flagged by fact-checkers; because every genuine article shares one source, the corpus is easy to separate and is widely suspected of source leakage. WELFake [3] merges four public corpora into 72k articles to reduce exactly this kind of overfitting and is our primary training set; one of its four sources is ISOT itself, and our preprocessing audit found that almost every ISOT article recurs verbatim in WELFake (Section III), which is why our cross-corpus experiment removes the shared articles from the WELFake side. FakeNewsNet [7] adds social context and PolitiFact/GossipCop labels to news items; we use its PolitiFact titles in our offline index. On the verification side, FEVER [8] contributes 185k claims written against Wikipedia and labelled Supports, Refutes or Not Enough Info, defining the retrieve-then-verify task that our stance module inherits. ClaimBuster [9] provides 23.5k debate sentences annotated for check-worthiness. Surveys by Zhou and Zafarani [10] and Guo et al. [11] organise this landscape: the former distinguishes knowledge-, style-, propagation- and source-based detection, and the latter formalises fact-checking as claim detection, evidence retrieval, verdict prediction and justification production—the sequence our pipeline follows.

B. Content-Based Classifiers

The earliest and still most common approach treats detection as document classification. Ahmed et al. [5] reported that TF-IDF n-grams with a linear SVM reach roughly 92 % accuracy on ISOT. Pre-trained transformers [12] raised these numbers further: FakeBERT [4] feeds BERT embeddings through parallel convolutional blocks and reports 98.9 % accuracy on a Kaggle corpus, and WELFake's own linguistic-feature model reports 96.7 %. DistilBERT [13] keeps about 97 % of BERT's language-understanding performance at 40 % fewer parameters, which is what makes fine-tuning on a 4 GB consumer GPU feasible and is why we adopt it. Blackledge and Atapour-Abarghouei [14] are a rare exception to in-domain-only evaluation: DistilBERT, BERT and DeBERTa reach 0.989–0.997 accuracy on an ISOT hold-out but only 0.670–0.697 when the ISOT-trained models are tested on another corpus. Apart from that study, what these papers share is an in-domain, binary evaluation without cross-corpus transfer, and none can tell a user why an article is suspicious beyond its wording. We keep such a classifier, but only as a secondary signal.

C. Claim Detection, Evidence Retrieval and Fact Verification

The ClaimBuster benchmark [9] is the standard training set for check-worthiness detection; our claim identifier combines similar hand-crafted cues with an optional classifier trained on that data. For verification, the FEVER baseline [8] chains document retrieval, sentence selection and an NLI model and scored 31.9 FEVER points, and KGAT [15] propagates evidence through a kernel-based graph attention network to exceed 70 FEVER points—the current ceiling for Wikipedia-grounded verification. DeClarE [16] verifies free-text claims against web search results with attention over the retrieved articles and a source-credibility embedding, which is close in spirit to our online path. Two components make retrieval and stance practical at our scale: Sentence-BERT [17] bi-encoders

(we use the 22M-parameter MiniLM variant [18]) turn semantic search over thousands of fact-checked statements into a millisecond nearest-neighbour lookup, while a small DeBERTaV3 cross-encoder [19] trained on SNLI and MultiNLI [20] provides an off-the-shelf three-way stance judgement on CPU.

D. Explainability

LIME [21] explains an individual prediction by fitting a sparse linear surrogate on perturbed inputs; it is model-agnostic and produces token-level highlights that lay users can read. In the fake-news setting, dEFEND [22] co-attends over article sentences and user comments so that the sentences and comments that drove the decision can be shown as the explanation, and DeClarE [16] exposes the retrieved evidence words its attention focused on. These methods explain a model; they do not by themselves ground the verdict in checkable facts. We therefore treat LIME highlights as an explanation of the style signal only, and build the primary explanation from the retrieved evidence, its stance, its source and its date.

E. Temporal and Contextual Misinformation

Much misinformation is not fabricated text but genuine material displaced in time or place. Vosoughi et al. [1] quantified how much faster and deeper falsehoods travel. In India, Al-Zaman [2] analysed 125 fact-checked COVID-19 stories and found that two-thirds concerned health, that text paired with video was the commonest format, and that almost all originated on social platforms rather than in mainstream media. Existing detectors ignore publication dates entirely; we add a lightweight check that compares the input's date and recency cues against the dates of the supporting evidence and against the entities it mentions.

F. The Gap

Table I makes the gap visible. Style-based classifiers are accurate in-domain but produce no evidence, no partial-truth labels and no abstention; FEVER-style verifiers ground their decisions in evidence but assume Wikipedia-only, synthetic claims and do not consider time; explainability work explains models rather than facts. No system we are aware of combines (i) a five-way verdict with an explicit Unverifiable class, (ii) evidence retrieval from free sources with an offline fallback, (iii) a temporal and contextual check, and (iv) evidence-linked explanations, while running on ordinary student hardware. Those properties are the three contributions listed in Section I, and the rest of this paper builds and evaluates them.

III. Datasets and Preprocessing

Table II lists the corpora and the role each plays. WELFake [3] (72,134 articles) is the primary training set for the content classifier and ISOT [5] (44,898 articles) the secondary one. LIAR [6] (12,836 PolitiFact statements, official splits) trains a three-way truthfulness head and,

TABLE I
Representative prior work compared on the properties this paper targets. Ev. = grounds the verdict in retrieved evidence; Exp. = explains the verdict to a user; Abst. = can abstain; Temp. = checks dates or context. Full table in docs/literature_table.md.

Work	Output	Ev.	Exp.	Abst.	Temp.
LIAR CNN [6]	6-way	–	–	–	–
ISOT SVM [5]	binary	–	–	–	–
WELFake [3]	binary	–	–	–	–
FakeBERT [4]	binary	–	–	–	–
FEVER baseline [8]	3-way	✓	–	NEI	–
KGAT [15]	3-way	✓	–	NEI	–
DeClarE [16]	binary	✓	✓	–	–
dEFEND [22]	binary	–	✓	–	–
This work	5-way	✓	✓	✓	✓

TABLE II
Datasets after preprocessing (make data).

Dataset	Rows	Labels	Role
WELFake	60,799	real / fake	classifier (primary)
ISOT	37,567	real / fake	leakage, cross-dom.
LIAR	12,836	6 → 3-way	LIAR head, index
FEVER subset	23,000	S / R / NEI	stance evaluation
FakeNewsNet (PF)	1,056	real / fake	offline index
Live Claims (ours)	95	5-class	end-to-end

together with 1,056 FakeNewsNet [7] PolitiFact titles, populates the offline fact-check index. A balanced FEVER [8] subset (20,000 train + 3,000 dev claims) evaluates the stance model. LIAR's six grades map onto our scheme as true/mostly-true → Real, half-true → Partially True, barely-true → Misleading and false/pants-fire → Fake; the three-way collapse used for the LIAR head is TRUE / MIXED / FALSE. Unverifiable has no counterpart in any corpus—it arises only from retrieval failure at run time.

Live Claims Set. For end-to-end evaluation we built 95 recent claims (fact-checks published between February 2025 and 25 October 2026, 93 of them in 2026) from PolitiFact (46), Snopes (31), Lead Stories (9), BOOM (4), Alt News (3) and Full Fact (2). Each rating, claim text and date was read from the live fact-check page and mapped to the five classes (e.g. Snopes Miscaptioned/Outdated → Misleading); Unverifiable comes only from Lead Stories evidence-absence ratings. Class counts are Fake 27, Real 21, Partially True 19, Misleading 19, Unverifiable 9. A stratified 30-item dev slice (seed 42) is reserved for tuning fusion thresholds; the remaining 65 items are held out.

Preprocessing. All corpora are converted to one schema (id, title, text, label, …, split). For WELFake and ISOT we (i) strip publisher and format boilerplate with 26 named text patterns and 2 title patterns—Reuters datelines, photo credits, captions, bylines, [VIDEO] tags, outlet suffixes, URL debris—while keeping content words such as "told Reuters"; (ii) drop empty and very short texts; (iii) de-duplicate greedily, first exact SHA-1 matches and then MinHash near-duplicates (Jaccard ≥ 0.9); and (iv)

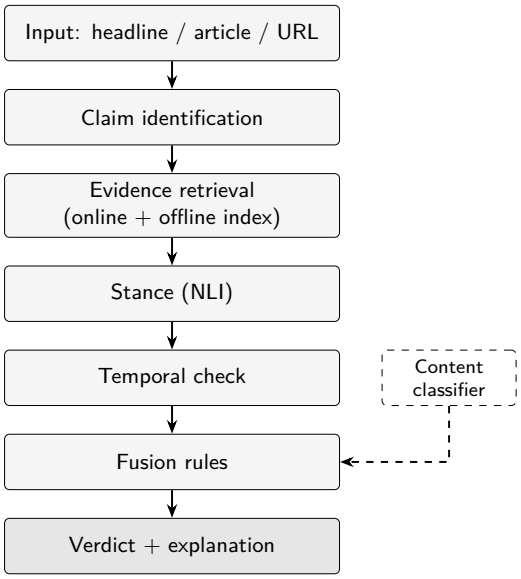


Fig. 1. System pipeline. Claims flow through evidence retrieval (online sources and the offline FAISS index), NLI stance and the temporal check into rule-based fusion. The content classifier (dashed) feeds fusion only as a confidence adjustment and the single-source tie-break R8.

split 80/10/10, stratified, seed 42, giving 48,639 / 6,080 / 6,080 WELFake and 30,053 / 3,757 / 3,757 ISOT rows. Splits are stored as id lists and every model reads the same ids.

Leakage audit: ISOT $\subset$ WELFake. WELFake was assembled from four corpora, one of which is ISOT. Hashing normalised text shows that 37,542 of the 37,567 processed ISOT articles (99.93 %) occur verbatim in processed WELFake, so "train on WELFake, test on ISOT" is an in-domain experiment. For the cross-domain experiment we therefore use WELFake\ISOT: the WELFake rows not hash-matched to ISOT, minus 25 MinHash near-duplicates, i.e. 23,232 rows. A second audit trains a logistic regression on the 26 artefact-pattern counts alone (no words): it reaches 0.9973 five-fold accuracy on raw ISOT but 0.5568 on processed ISOT, which confirms that raw ISOT can be separated by source boilerplate alone and that our stripping removes that shortcut.

IV. Methodology

Fig. 1 shows the pipeline; every box is a module under src/ and runs behind one FastAPI endpoint, POST /analyze, in an online and an offline mode.

A. Claim Identification

The input is cleaned and sentence-split with spaCy en_core_web_sm. Sentences of fewer than six tokens, questions and imperatives are dropped; each remaining sentence s at position i receives the check-worthiness score

$$h(s) = 1.0\,[\mathrm{NE}] + 0.8\,[\mathrm{NUM}] + 0.6\,[\mathrm{REP}] + 0.5\,[i < 3] + 0.4\,[\mathrm{CMP}] - 0.8\,[\mathrm{OPN}] - 0.5\,[\mathrm{HDG}], \quad (1)$$

where the indicators denote a PERSON/ORG/GPE/EVENT/NORP/LOC entity, a CARDINAL/PERCENT/MONEY/DATE/QUANTITY entity, a reporting verb (said, announced, according to, …), a lead sentence, a number or comparative, a first-person opinion cue (I think, should, must) and a hedge (may, might, could). The score is normalised to $[0, 1]$ by its maximum (3.3) and clipped. Near-duplicate sentences (MiniLM cosine > 0.85) are removed and the top $k = 5$ are kept; a headline is always claim 1 with score 1. The optional ClaimBuster-trained classifier [9] is not used yet.

B. Evidence Retrieval

Online, each claim is sent to DuckDuckGo text search (ddgs), the Wikipedia search API (top page intros) and, when an API key is configured, the Google Fact Check Tools claims:search endpoint, each with a 6 s timeout. Result pages are fetched and extracted with trafilatura, dated with htmldate, and reduced to the three sentences most similar to the claim. Result pages of Tier-1 fact-checkers are also parsed for their published rating (schema.org ClaimReview, or PolitiFact's Truth-O-Meter), so that a fact-check found by web search carries its rating like an index item. All network answers are cached in SQLite so that runs are repeatable. Offline, the claim is embedded with all-MiniLM-L6-v2 [17], [18] and searched in a FAISS [23] IndexFlatIP over L2-normalised vectors (cosine) of all 12,836 LIAR statements and 1,056 FakeNewsNet titles (13,892 vectors, 20.35 MB); hits with cosine < 0.55 are discarded. Each item receives a source-reliability weight w_s from an editable domain list: fact-checkers 1.0, wire services and official sites 0.8, Wikipedia 0.6, unknown domains 0.3, satire/hoax sites 0 (flagged).

C. Stance Detection

Each (claim, evidence) pair is scored by the cross-encoder cross-encoder/nli-deberta-v3-small, a DeBERTaV3 [19] model trained on SNLI [24] and MultiNLI [20] only; we do not fine-tune it. With premise = evidence and hypothesis = claim, entailment, contradiction and neutral map to Supports, Refutes and Neutral with probability p. For items that carry a fact-checker rating (LIAR, FakeNewsNet, Google Fact Check, parsed ClaimReview) the rating is turned into a stance with $p = 0.95$, but only when the claim restates the rated statement (NLI entailment, or cosine ≥ 0.75 without contradiction). The signed contribution of an item is $\sigma(e) = \mathrm{sign}(\mathrm{stance}) \cdot p \cdot w_s$.

D. Temporal and Context Check

Three flags are computed for the main claim. Old news: the input contains a recency cue (breaking, just in, today, …) and the gap between the input date t_{post} and the (lower) median date of the supporting evidence is at least $N = 6$ months; or the claim names its own date and that date lies at least N months after the supporting

TABLE III
Fusion rules (src/fusion/rules.py); first match wins. "Imp." = another claim with score $c_i \geq 0.6$.

Rule	Condition	Verdict
LANG	input is not English	Unv.
R0	input domain on the satire / hoax list	Fake
R1	no evidence, all evidence neutral, or $E_1 = 0$	Unv.
R2	Tier-1 fact-check rated false, stance Refutes, cosine to main claim ≥ 0.75	Fake
R3	main Refutes and $E_1 \geq 3$	Fake
R4	main Supports and a temporal/context flag	Misl.
R5	main Supports, an imp. claim has $S_i \leq -\tau$, $E_i \geq 3$	Part.
R6	main Neutral, imp. claims split (one $S_i \geq \tau$, one $S_i \leq -\tau$, each $E_i \geq 3$)	Part.
R7	main Supports and $E_1 \geq 3$	Real
R8	main non-neutral and $E_1 = 1$: Fake if Refutes and $p_{\text{fake}} \geq 0.85$; Real if Supports and $p_{\text{fake}} \leq 0.15$; else	Unv.
R9	anything else	Unv.

evidence. Evidence without a date is not used. Out of context: the claim's place (GPE) or year entities conflict with those of at least two supporting passages. Headline–body mismatch: NLI with the body as premise refutes the headline with $p \geq 0.7$.

E. Verdict Fusion and Confidence

For claim i let $S_i = \sum_e \sigma(e)$ and let E_i be the number of non-neutral items with $w_s \geq 0.5$. The main stance is Supports if $S_1 \geq \tau$, Refutes if $S_1 \leq -\tau$ and Neutral otherwise, with $\tau = 0.8$. Table III lists the decision rules, evaluated top to bottom (first match wins). The thresholds are those of src/fusion/fusion.yaml: $\tau = 0.8$, minimum evidence $m = 3$, $N = 6$, $k = 5$. They were selected by a grid search over τ, m and N on the 30 dev claims and the 15 temporal-set controls only, with the selection rule fixed in advance; this moved τ from 0.6 to 0.8 and m from 2 to 3 and raised dev macro-F1 from 0.0857 to 0.1165, i.e. by one claim, so it is a sanity check rather than a precise optimum. The content classifier's fake probability p_{fake} appears in only two places: the R8 tie-break and a ± 0.10 confidence adjustment.

The confidence combines how one-sided, how broad and how reliable the evidence for the main claim is:

$$a = |S_1| \,/\, \left(\sum_e p_e w_e + 10^{-6}\right), \quad c = \min(E_1, 4)/4, \tag{2}$$

$$q = \text{mean}(w_s \text{ over non-neutral items}), \tag{3}$$

$$\text{conf} = \text{clip}(0.5a + 0.3c + 0.2q + \delta,\ 0.05,\ 0.98), \tag{4}$$

where $\delta = +0.10$ if the classifier agrees with a Fake/Real verdict ($p_{\text{fake}} \geq 0.7$ / ≤ 0.3), -0.10 if it disagrees and 0 otherwise. For Unverifiable, conf $= 1 - \max(a, 0.2)$, i.e. how sure the system is that it cannot tell. R0 and the language check use fixed confidences of 0.9 and 0.8. Every response returns the rule that fired, the claims, the evidence cards with source, date, cosine and stance, and the temporal flags.

TABLE IV
Content classifiers, test split (in-domain). Val = validation macro-F1; DB = DistilBERT (full train, max_len 256).

Data	Model	Val	Acc	P	R	F1	AUC
WELFake	NB	0.8728	0.8771	0.8744	0.8782	0.8758	0.9443
	LR	0.9455	0.9462	0.9457	0.9449	0.9453	0.9875
	SVM	0.9495	0.9490	0.9485	0.9478	0.9481	0.9884
	DB	0.9930	0.9900	0.9905	0.9891	0.9898	0.9995
ISOT	NB	0.9550	0.9470	0.9458	0.9472	0.9464	0.9865
	LR	0.9892	0.9848	0.9854	0.9839	0.9846	0.9986
	SVM	0.9900	0.9878	0.9882	0.9870	0.9876	0.9990
	DB	–	0.9992	0.9992	0.9992	0.9992	1.0000

F. Content Classifiers

We train TF-IDF word n-gram models with multinomial Naive Bayes, logistic regression and a linear SVM (scikit-learn), and fine-tune distilbert-base-uncased [13], [25] on title [SEP] text. A second DistilBERT head is fine-tuned on LIAR statements (text only, no speaker or party, which are label proxies) for TRUE / MIXED / FALSE with class-balanced cross-entropy. The deployed style signal is the full-data DistilBERT checkpoint. WELFake and ISOT use a stratified 80/10/10 split (seed 42) and LIAR its official split; hyper-parameters are chosen by validation macro-F1 (5-fold grid search for the classical models), and all training ran on one laptop GPU (RTX 2050, 4 GB).

V. Results

A. Content Classifiers (In-Domain)

Table IV gives held-out test results. Among the tuned classical models the linear SVM is best on WELFake (macro-F1 0.9481, ROC-AUC 0.9884), logistic regression is within 0.003 and Naive Bayes trails by about 0.07; validation–test gaps are at most 0.0086, so the grid search did not over-fit the validation split. The fine-tuned DistilBERT is clearly better: 0.9898 macro-F1, 0.9900 accuracy and 0.9995 ROC-AUC on the 6,080 WELFake test articles (Fig. 2, Fig. 3), which meets objective O1 (macro-F1 ≥ 0.95). ISOT scores (up to 0.9992) are higher still, but ISOT is in-domain here—almost all of it is contained in WELFake—and single-source on the real side.

B. Cross-Domain Transfer and Artefacts

Table V trains on one domain and tests on the other, using the disjoint WELFake\ISOT domain. Leaving the training domain costs 0.29–0.30 macro-F1 from WELFake\ISOT to ISOT and 0.43–0.44 in the other direction, so the near-perfect in-domain numbers say little about unseen sources. Evaluated on raw text with the artefacts kept, transfer looks better (e.g. 0.7616 instead of 0.6638 for LR, W\I → ISOT), i.e. part of what "transfers" is boilerplate. The cross-domain experiment was run for the classical models only; the DistilBERT cross-domain

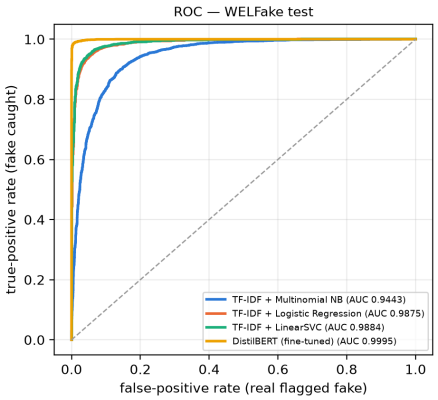


Fig. 2. ROC curves of the classifiers on the WELFake test split.

TABLE V
Cross-domain macro-F1, test splits (rows = train, columns = test). Proc. = artefacts stripped; raw = artefacts kept.

		Processed		Raw	
Model	Train	W\I	ISOT	W\I	ISOT
LR	W\I	0.9583	0.6638	0.9645	0.7616
	ISOT	0.5502	0.9846	0.5663	0.9933
SVM	W\I	0.9601	0.6611	0.9676	0.7611
	ISOT	0.5480	0.9876	0.5601	0.9949

cells would need two further GPU trainings and are not reported. This drop is the empirical reason that, in our system, evidence and not the classifier decides the verdict.

C. DistilBERT: Sweep and Full Training

Table VI lists every DistilBERT run. On a 20k-row training subset the three learning rates at length 256 lie within 0.0008 validation macro-F1 (0.9863–0.9871); 5×10^{-5} was best by 0.0002 over 2×10^{-5}, a near-tie, and its validation loss rose from epoch 2 (0.0572) to 3 (0.0671) on the subset. Retrained on the full 48,639-article training split with that setting, validation macro-F1 rose every epoch (0.9870, 0.9906, 0.9930) and validation loss fell every epoch, so the subset over-fitting did not recur; training took 31.6 minutes with 1.68 GiB peak GPU memory. Of the 6,080 test articles 61 are misclassified (47 missed fakes, 14 false alarms; Fig. 3). The length-512 arm of the sweep was not run, so the effect of longer inputs remains open; 27 of the 61 errors are articles longer than twice the 256-token window.

Classifier error analysis. The 61 DistilBERT errors were assigned to categories by fixed, ordered rules: 27 truncation (the article is more than twice the input window), 17 style overlap (wire-style fakes or tabloid-style real articles), 6 opinion or first-person commentary, 5 source cues pointing the wrong way, 4 short texts and 2 wire copy republished on fake-labelled sites. Examples: welfake_8702, a fabricated story about an Israeli minister and Italian earthquakes written in news-agency register,

TABLE VI
DistilBERT on WELFake (3 epochs, best epoch 3 in all runs). Sweep runs use a 20k-row training subset; all use the full validation split.

Run (lr)	max_len	val macro-F1	val acc	min
sweep 2×10^{-5}	256	0.9869	0.9872	14.8
sweep 3×10^{-5}	256	0.9863	0.9865	14.3
sweep 5×10^{-5}	256	0.9871	0.9873	15.2
sweep $\{2, 3, 5\} \times 10^{-5}$	512	not run		
full train 5×10^{-5}	256	0.9930	0.9931	31.6

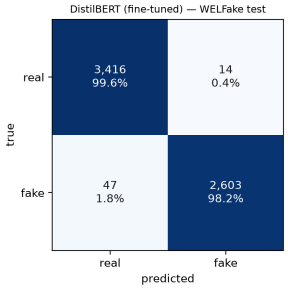


Fig. 3. DistilBERT confusion matrix on the WELFake test split (rows = gold): 3,416 / 14 real, 47 / 2,603 fake.

receives $P(\text{fake}) = 0.000$; welfake_9677, an 887-word fake about Saturn's hexagon, is classified real from its first 256 tokens. These are errors of style and length, not of fact.

D. LIAR Truthfulness Head

The best configuration (lr 3×10^{-5}, 3 epochs, best epoch 2, validation macro-F1 0.4773) reaches 0.4669 test accuracy and 0.4667 macro-F1 on the three-way task, against 0.3749 for the majority class (Table VII). Its six-way twin with the same settings reaches 0.2689 accuracy, on par with Wang's text-only CNN (0.270) [6]: short statements carry little truth signal in their wording, which is why the head is only an auxiliary signal.

E. Stance Detection on FEVER

On 3,000 balanced FEVER dev pairs (gold evidence text from the FEVER 2017 Wikipedia dump; NEI claims paired with the sentences retrieved for them by a FEVER shared-task retrieval system, as distributed with the gold-evidence release; premise prefixed with the page title) the cross-encoder reaches 0.7267 accuracy and 0.7121 macro-F1 (Table VIII, Fig. 4). This is a zero-shot result—the model was trained on SNLI and MultiNLI only, not on FEVER—and it misses our objective O4 (≥ 0.85). The shortfall is concentrated in NEI: on the gold Supports/Refutes pairs accuracy is 0.8790, but 479 of 1,000 NEI pairs are predicted Refutes, because the retrieved sentences concern the same entity and state a different fact. Without the page-title prefix accuracy falls to 0.6337. In the pipeline this error mode is damped by R3, which needs three reliable refuting sources.

TABLE VII
LIAR official test split (1,283 statements).

Model	Labels	Acc	Macro-F1
Majority class	3	0.3749	–
DistilBERT 3-way head (ours)	3	0.4669	0.4667
Majority class	6	0.2081	–
DistilBERT 6-way twin (ours)	6	0.2689	0.2686
Wang (2017) text-only CNN	6	0.270	n/r
Wang (2017) hybrid CNN	6	0.274	n/r

TABLE VIII
Zero-shot stance on 3,000 balanced FEVER dev pairs.

Class	Precision	Recall	F1
Supports	0.8488	0.8870	0.8675
Refutes	0.6190	0.8710	0.7237
Neutral (NEI)	0.7701	0.4220	0.5452
Accuracy / macro-F1	0.7267 / 0.7121		

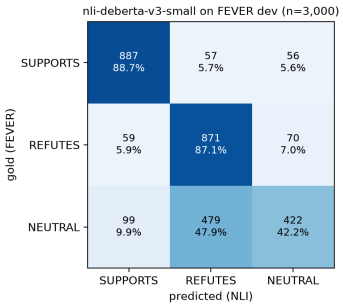


Fig. 4. Stance confusion matrix on the FEVER dev subset (rows = gold).

TABLE IX
End-to-end five-class results on the 65 held-out Live Claims (v2 = with the leakage guard; text-only = 58 claims not about an image or video).

Run	Acc	Macro-F1	Unv. P
v2 online	0.2000	0.1667	0.0943
v2 online, text-only	0.1724	0.1463	0.0816
v2 offline	0.0923	0.0343	0.0938
v1 online, leaky (no guard)	0.2308	0.1854	0.0789
v1, own page removed post hoc	0.1538	0.1260	0.0909
Target (O6)	–	0.55	0.70

F. End-to-End Evaluation on Live Claims

Each of the 65 held-out Live Claims is sent through the same code as POST /analyze as a headline dated with its claim date; the gold label is the fact-checker's rating mapped to five classes. Online retrieval used DuckDuckGo, Wikipedia and the offline index (no Google Fact Check key was configured). Table IX gives the results and Table X the per-class breakdown. Online, the system reaches 0.2000 accuracy and 0.1667 macro-F1; offline, where the LIAR/FakeNewsNet index gives at least one hit with cosine ≥ 0.55 for only 36.9 % of the claims, it reaches 0.0923 and 0.0343. Objective O6 (macro-F1 ≥ 0.55, Unverifiable precision ≥ 0.70) is not met; objective O3 (at least one online passage for ≥ 85 % of claims) is met with 100 %. The errors have one dominant shape: 53 of 65 answers are Unverifiable, Partially True and Misleading are never predicted, and when the system does commit, it is mostly right (Real and Fake precision 0.6667 each). Rules R1 (29) and R9 (18) fired most often, R3 six times and R2 three times.

A leakage finding. Our first end-to-end run (v1) scored 0.2308 accuracy and 0.1854 macro-F1. Inspecting its evidence showed that web search often returned the claim's own fact-check page—the very page the gold label was taken from—and that for 34 of the 65 claims this page was among the kept evidence (a lower bound, matched on normalised URLs). Removing those items after the fact and re-fusing lowers v1 to 0.1538 accuracy and 0.1260 macro-F1, the honest v1 baseline. We therefore added an evaluation-only guard that drops evidence whose normalised or canonical URL equals the claim's fact-check URL; in v2 it removed the own page for 35 of 65 claims, while 22 claims kept at least one other matching fact-check, which is legitimate evidence. In deployment, finding an existing fact-check is the intended behaviour; in evaluation it silently turns the task into label lookup. Any evaluation of retrieval-based verification on claims taken from fact-checkers needs such a guard.

G. Ablation

Table XI switches components off. Evidence and stance improve macro-F1 over the style classifier alone (0.1667 vs. 0.1328) but not accuracy (0.2000 vs. 0.2462): the classifier can only answer Real or Fake and is right on a quarter of these claims by doing so, whereas the evidence path abstains. The temporal check changes no verdict, so the expected ordering (classifier $<$ +evidence $<$ +temporal) does not hold for the temporal step. Offline mode is the weakest variant because the index rarely covers 2025–26 claims.

H. Temporal Test Set

We built a separate set of 30 verified fact-checks: 15 items of old material recirculated as new and 15 current controls. Run online, the old-news flag never fired: 0 true positives, 0 false positives, 15 false negatives and 15 true negatives, i.e. 50 % correct, so objective O5 (≥ 80 %) is not met. Of the 15 missed items, 7 had no dated supporting evidence, 3 had dated evidence but neither a recency cue nor a date in the claim, and 5 had both but a gap below $N = 6$ months, because the supporting pages were recent re-reports or fact-checks of the old material. Fact-check phrasings rarely contain "breaking" or "today", and the dated evidence the check needs is precisely what retrieval of a recirculated story does not return.

TABLE X
Per-class results, v2 online (65 claims). Pred. = number of times the class was predicted.

Class	P	R	F1	Gold	Pred.
Real	0.6667	0.1429	0.2353	14	3
Fake	0.6667	0.3158	0.4286	19	9
Partially True	0.0000	0.0000	0.0000	13	0
Misleading	0.0000	0.0000	0.0000	13	0
Unverifiable	0.0943	0.8333	0.1695	6	53

TABLE XI
Ablation on the 65 held-out Live Claims (v2).

Variant	Acc	Macro-F1	F1 Real	F1 Fake
(a) classifier only	0.2462	0.1328	0.264	0.400
(b) + evidence & stance	0.2000	0.1667	0.235	0.429
(c) + temporal (full)	0.2000	0.1667	0.235	0.429
(d) full, offline only	0.0923	0.0343	0.000	0.000

I. Latency and Resources

On the laptop CPU (8 threads; other jobs running, so the numbers are pessimistic) the median end-to-end time per claim is 40 ms offline (p90 164 ms), 2,890 ms online with a warm cache (p90 4,611 ms) and 11,032 ms online when pages must be fetched (p90 15,869 ms; median retrieval stage 9,683 ms, measured on the first evaluation pass); the median stance stage takes 1,154 ms online with a warm cache. With all models loaded (MiniLM, the NLI cross-encoder, spaCy and DistilBERT) the process peaks at 2,373.8 MB resident memory, within our 4 GB CPU budget. The system ships as a web interface (analyse, result and about screens with evidence cards, a dates panel, LIME highlights and JSON download) served from one CPU-only Docker image of about 3.1 GB with all models baked in. Run offline with networking disabled, the container answered all five gallery examples with their intended verdicts at a median of 0.50 s (0.22 s on repeat) and a peak resident memory of 1.33 GB.

J. Case Studies and Failure Analysis

Real outputs illustrate both the strengths and the failure modes. "WHO declared that 5G towers spread COVID-19." is labelled Fake by R3: four reliable sources refute it (five refuting, five neutral passages). "Michelle Obama is a man." (LC020) is Fake by R3 with confidence 0.98. The true headline "ISRO's Chandrayaan-3 landed near the lunar south pole on 23 Aug 2023" ends in Unverifiable (R9, one supporting, three refuting, six neutral passages) while DistilBERT gives it $P(\text{fake}) = 0.9797$; a classifier-first system would have called a true headline fake. LC003, a true claim that U.S. families paid more than $1,700 each in tariff costs, ends in Unverifiable (R1) although a Joint Economic Committee passage restating it was retrieved (cosine 0.85): the NLI model labelled all ten passages neutral. The hyper-local "A local shop in Ghaziabad sold 500 phones in one hour" is correctly Unverifiable in v2 but was Fake by R3 in v1: two reliable passages about a phone theft in Ghaziabad were read as refuting—the not-enough-information-to-Refutes confusion measured on FEVER—and only the stricter $m = 3$ rule now stops them. Sixteen malformed or adversarial inputs (empty, emoji-only, Hindi, 60k-character, HTML injection, URL in offline mode, …) produced no server error.

VI. Discussion

A. Comparison with Existing Systems

Table XII places the system among manual fact-checkers, fact-check search tools, style classifiers and evidence-based verifiers. Published numbers are copied from the cited papers; they come from other data and splits and are not directly comparable with ours, so the last column gives our closest measured counterpart.

Where existing systems beat us. Manual fact-checkers are far more accurate: our labels come from them, and the system matches their rating on only 13 of the 65 held-out claims (accuracy 0.2000). Binary style classifiers report higher accuracy on their own benchmarks—and so does our DistilBERT at 0.9898 macro-F1—but those numbers measure the style of one corpus; across disjoint corpora our classifiers lose 0.29–0.44 macro-F1 and the DistilBERT models of [14] fall from 0.989 to 0.670 accuracy. FEVER verifiers such as KGAT are much stronger on Wikipedia claims, where retrieval targets a closed, well-formed corpus; our untuned stance model alone misses 0.85 even with gold evidence. When a matching fact-check exists, a fact-check search tool returns a professional rating, which is more reliable than our NLI over news passages.

What we do that they cannot. (i) A verdict when no fact-check exists: R3 and R7 decide from ordinary news and reference sources (e.g. the 5G case), which a fact-check search tool cannot. (ii) A date check: no system in Table XII compares the dates of a claim and its evidence; ours can turn a supported but outdated claim into Misleading, although on our temporal set it has not yet fired. (iii) Honest abstention: Unverifiable is a first-class answer with an uncertainty sentence, and every verdict names the rule and the evidence behind it. Our abstention is currently too frequent (precision 0.0943), but when the system does commit to Real or Fake its precision is 0.6667 for each.

B. Limitations

1) Low end-to-end performance. Five-class macro-F1 is 0.1667 against a target of 0.55; 53 of 65 answers are Unverifiable and Partially True and Misleading are never predicted.
2) Small evaluation set, single labelling. 65 held-out claims (95 in all) with labels mapped from one fact-checker's rating; no second annotator or agreement statistic, and the fusion thresholds were tuned on only 30 claims.

TABLE XII
Comparison with existing systems. Ev. = verdict grounded in retrieved evidence; Temp. = checks dates; Abst. = can abstain. Published results are those reported in the cited work (other data and splits); our numbers are from Section V.

System	Family	Output	Ev.	Temp.	Abst.	Published result	Ours (closest comparable)
PolitiFact / Snopes / BOOM	manual	rating scale	✓	✓	✓	gold standard; source of our labels	13 of 65 of their ratings matched (acc. 0.2000, macro-F1 0.1667)
Google Fact Check Explorer	search tool	existing ratings	links	–	–	returns only claims already checked	also decides from news evidence when no check exists (R3/R7)
LIAR CNN [6]	style	6-way	–	–	–	0.270 acc. (text only)	6-way twin 0.2689 acc.
ISOT SVM [5]	style	binary	–	–	–	≈92 % acc. on ISOT	SVM 0.9876 F1 in-domain, 0.5480 cross-domain; artefacts alone 0.9973 on raw ISOT
WELFake [3]	style	binary	–	–	–	96.7 % acc. on WELFake	DistilBERT 0.9900 acc. (boiler-plate stripped)
FakeBERT [4]	style	binary	–	–	–	98.9 % acc. (Kaggle corpus)	not evaluated on that corpus
DistilBERT [14]	style	binary	–	–	–	0.989 acc. on ISOT, 0.670 on another corpus	SVM 0.9601 → 0.6611 F1 (W\I → ISOT)
FEVER baseline [8]	evidence	3-way	✓	–	NEI	31.9 FEVER score	stance only: 0.7267 acc. on gold evidence (zero-shot)
KGAT [15]	evidence	3-way	✓	–	NEI	>70 FEVER score	same; no FEVER score computed
DeClarE [16]	evidence	credibility	✓	–	–	binary label, no abstention	five classes incl. Unverifiable
This work	evidence	5-way	✓	✓	✓	–	macro-F1 0.1667 online, 0.0343 offline

3) Retrieval dependence. Results depend on an unofficial DuckDuckGo interface and on what the web returns on the day; no Google Fact Check key was used, and the offline index covers only 36.9 % of recent claims.
4) Zero-shot stance. The NLI model reaches 0.7267 on FEVER, confuses not-enough-information with Refutes, and labelled a near-verbatim restatement neutral (LC003).
5) Temporal check unproven. It fired on none of the 15 recirculated items, because the dated supporting evidence it needs is rarely retrieved.
6) Scope. English and text only; 7 of the 65 held-out claims concern an image or a video.
7) Classifier evidence incomplete. DistilBERT was not trained at length 512 nor evaluated cross-domain; the source-reliability tiers are hand-made; and the leakage guard matches URLs only, so it is a lower bound on what it removes.

VII. Conclusion and Future Work

We presented an evidence-grounded, explainable pipeline that verifies a headline, article or URL into one of five verdicts—Real, Fake, Partially True, Misleading or Unverifiable—using only free tools and a student laptop. Claims are extracted, matched against free online sources and an offline FAISS index of fact-checked statements, labelled for stance by an NLI cross-encoder, checked for temporal and contextual mismatch, and fused by a transparent decision table that reports the rule that fired. The content classifier may only adjust confidence or break a single-source tie.

The style classifiers are strong in-domain—DistilBERT reaches 0.9898 test macro-F1 on WELFake, the tuned SVM 0.9481—but this does not transfer: trained on WELFake\ISOT and tested on ISOT the SVM falls to 0.6611, in the reverse direction to 0.5480, and an artefact-only classifier separates raw ISOT at 0.9973. On 65 recent held-out fact-checked claims the full system reaches 0.2000 accuracy and 0.1667 five-class macro-F1 online and 0.0343 offline, well below our target; the zero-shot stance model reaches 0.7267 on FEVER and the old-news check did not fire on our temporal set. We also found that our own first evaluation leaked the gold fact-check page into the evidence for 34 of 65 claims and over-stated macro-F1 (0.1854 instead of 0.1260); we report the guarded numbers and recommend such a guard for any evaluation that draws claims from fact-checkers.

Future work follows from these results: fine-tune the stance model on not-enough-information pairs, query the Google Fact Check Tools API, use the claim's own year and dated evidence in the temporal check, learn the fusion thresholds from a larger, double-labelled claim set, complete the length-512 and cross-domain DistilBERT runs, and add Hindi/Hinglish input and OCR for image-based claims.

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